

BEAM3R: Beam’s-eye-view architecture with Mamba-3 for implicit dose reconstruction

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Abstract. To enable accurate and rapid photon control point and proton beamlet dose calculation in the DoseRAD2026 challenge, our team presents BEAM3R, a dose estimation framework operating in beam’s-eye-view (BEV). Our core innovation combines a Mamba-3 state-space depth-sequence core and physics-based transport conditioning to accurately model long-range depth transport without computationally expensive 3D convolutions. BEAM3R shares a core 2D CNN encoder-decoder architecture for photon (Tasks 1 and 2) and proton (Tasks 3 and 4) dose tasks to process per-plane BEV slices. For proton beamlets, the network was conditioned by calculating and providing the water equivalent thickness and remaining range as inputs, so that the parameters determining Bragg peak position were readily available. For sequence modelling, photon models utilise a bidirectional Mamba-3 core to capture dose contributions from materials physically downstream from the current calculation point, whereas the proton model deploys a forward Mamba-3 core with learned energy-prefix tokens and a Bragg-peak refinement module. To reduce interpolation artifacts and support high spatial resolution estimation, we introduce axial grid alignment of BEV lattices with CT slices, and employ an implicit super-resolution representation via sub-pixel phase packing of inputs along with a differentiable Triton-accelerated resampler that evaluates predicted packed cubic B-spline coefficients directly in CT space. For MRI dose prediction tasks (Tasks 2 and 4), synthetic CTs (sCT) were generated from a patch-based conditional GAN with a SwinUNETR backbone, trained with least squares adversarial and MSE losses. Dose models were trained using bfloat16 precision and Adam optimiser with a learning rate of 1×10^{-4} to 2.5×10^{-4} using a plateau scheduler. The loss function combined MSE and a 10% dose threshold masked MAE term and an additional non-negativity constraint for protons. On the preliminary DoseRad2026 test set, CT-to-photon and CT-to-proton models achieved 1%/1 mm local gamma pass rates of 96.8% and 96.0%, respectively, with stratified plan-level MAEs of 0.0041 and 0.0079. Substitution of sCT reduced the corresponding gamma pass rates to 89.7% for photons and 75.4% for protons, with plan-level MAEs of 0.0093 and 0.0336, respectively. Standardised run-times were 23.4 s and 22.5 s for CT-to-photon and CT-to-proton prediction, increasing to 39.7 s and 42.8 s for the corresponding MRI-based pipelines.

Keywords: deep learning · proton dose calculation · photon dose calculation · synthetic CT · Mamba-3.

1 Introduction

Monte Carlo (MC) simulation is the gold standard for radiotherapy dose calculation [16], but its computational cost limits its use in time-sensitive applications such as online adaptive radiotherapy [10] and dose-guided radiotherapy [4]. The DoseRAD2026 Grand Challenge benchmarks rapid photon and proton beam-level dose calculation from CT and MRI. Beam-level modelling enables learning of beam-to-anatomy dose relationships, supporting flexible plan optimisation and dose accumulation workflows [6].

Beam’s-eye-view (BEV) slice-sequence models represent dose transport along a common beam-depth axis, reducing the need to encode treatment geometry explicitly while supporting efficient beam-level prediction [13, 14, 22]. Following this approach, we formulate 3D dose calculation using a 2D CNN encoder, a depth-sequence core, and a 2D CNN decoder.

We build on the BEV CNN-sequence architecture of Xiao et al. [22] and introduce targeted modifications to improve beam coverage, BEV-to-CT resampling accuracy, model capacity, and downstream depth context. Photon and proton models share this framework but use task-specific sequence cores and inputs, including proton transport conditioning. For the MRI tasks, a SwinUNETR-based [2] conditional generative adversarial network first generates a synthetic CT (sCT), which is passed to the corresponding CT-to-dose model. We apply these methods to all four challenge tasks: CT- and MRI-based photon dose calculation and CT- and MRI-based proton dose calculation.

2 Methods

2.1 Data

Separate models were trained for each task and anatomical cohort using only the DoseRAD2026 training dataset [21]. Table 1 summarises the patient-level splits used by both tasks. Patient-level splitting ensured that no control point or beamlet from a validation patient appeared in training. Hyperparameters and checkpoints were selected using the held-out validation cohort. s

2.2 Model

Architecture overview. Figure 1 shows the architecture design used for our dose prediction models. Our models build on the BEV CNN-sequence architecture of Xiao et al. [22], retaining its strided encoder and decoder, residual depthwise-pointwise spatial mixing, and residual sequence blocks. The photon model predicts the dose from one control point, whereas the proton model predicts the dose from one beamlet. Two complementary measures reduce BEV-to-CT resampling error: lattice snapping aligns one BEV axis with CT slices

Table 1. Patient-level training and validation splits.

Task/modality	Cohort	Training patients	Validation patients	Split seed
Photon	Abdomen	33	1ABB118, 1ABB164, 1ABB169	333
Photon	Thorax	36	1THB017, 1THB023, 1THB121	333
Proton	Abdomen	34	1ABB030, 1ABB083	42
Proton	Thorax	37	1THB095, 1THB211	42

at input, while packed spline coefficients provide a finer implicit output field without materialising a dense BEV volume. A per-plane CNN encoder extracts spatial features from BEV slices, a sequence core propagates information along beam depth, and a per-plane decoder predicts packed cubic B-spline coefficients. A differentiable resampler evaluates the resulting continuous dose field on the patient CT grid.

BEV geometry and inputs. BEV coordinates align dose transport with a common depth axis and avoid explicitly encoding the gantry angle, isocentre, and beam direction [13, 14, 22]. The photon BEV depth was increased from the 51.2 cm extent used by Xiao et al. [22] to 76.8 cm to avoid truncating long thoracic and abdominal beam paths. Photon inputs comprised of the normalised CT and projected MLC aperture, while proton inputs consisted of the normalised CT, Gaussian spot fluence, cumulative water-equivalent thickness (WET), and remaining range. The BEV lattice was snapped to CT slice positions along the axial axis, eliminating interpolation in that direction. Fine-lattice samples were phase-packed as channels to increase effective input resolution without increasing the encoder grid or depth-sequence length.

Proton model physics conditioning. Our key innovation is that we supply derived transport quantities to the network. To encode the physical determinant of Bragg-peak position directly, the proton model was conditioned on cumulative WET and remaining range along each ray. WET was calculated from an energy-dependent relative stopping-power estimate as

$$\text{WET}(d) = \int_0^d \text{RSP}(s) \, ds. \quad (1)$$

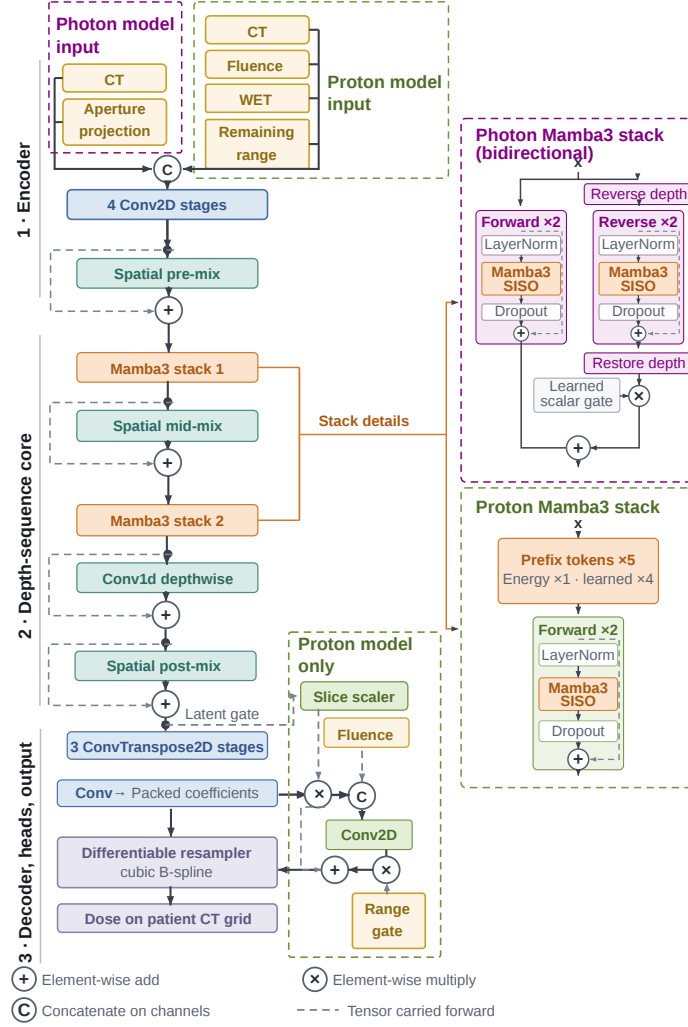
and the remaining range

$$R_{\text{rem}}(d) = R(E) - \text{WET}(d). \quad (2)$$

was calculated using the CSDA range approximation of Bortfeld [1] over the dataset energy range of 31.729–200.797 MeV spanned by the dataset.

$$R(E) = 0.022 E^{1.77} \text{ mm}, \quad (3)$$

The input also included a Gaussian fluence map parameterised by the beamlet spot width. The spot width σ is taken from the beamlet metadata where present and otherwise from the machine’s energy–spot table, and enters as



Tensor dimensions (D × H × W)

MODEL	PACKED INPUT	C	LATENT	PACKED OUTPUT
Proton	512×128×48 · 16 ch	96	512×16×6×96	512×128×48 · 4 ch
Mamba-3	implicit: 1024×256×48	3 heads	D × H × W × C	implicit: 1024×256×48
Photon	384×200×200 · 8 ch	128	384×25×25×128	384×200×200 · 4 ch
Mamba-3	implicit: 768×400×200	4 heads	D × H × W × C	implicit: 768×400×200

Fig. 1. Left: Shared BEV architecture for photon and proton dose prediction. Green dashed boxes mark proton-only components and the magenta dashed box shows the photon-only components.

$\exp[-(y^2 + z^2)/(2\sigma^2)]$ about the central ray. Gaussian tails falling outside the cuboid are truncated to zero without renormalisation; with a maximum σ of

7.99 mm against a 128 mm transverse extent. These continuous transport quantities directly encode the expected Bragg-peak location and were computed for each beamlet.

Encoder. The encoder comprises of four convolutional layers, with three strided layers providing eightfold in-plane downsampling. It maps the phase-packed inputs to a latent feature grid, using LeakyReLU activations after each convolution. A residual spatial mixing block, consisting of a depthwise 7×7 convolution followed by pointwise 1×1 channel mixing, refines the encoded features. The per-slice latent representation is then rearranged into depth-ordered sequences so that each spatial position forms an independent feature sequence ordered along beam depth.

Sequence Core. The recurrent core mixes information along depth in the coarse grid latent space. Both proton and photon dose models use single-input-single-output (SISO) Mamba-3 blocks [7], chosen for accuracy without the slower training throughput of the multi-input, multi-output (MIMO) variant. Mamba-3 uses a complex-valued exponential-trapezoidal state-space recurrence, which provides stronger state tracking and associative recall than the previous iteration (Mamba-2), keeps a fixed-size recurrent state and linear-time cost, and is deployed with memory-efficient chunked training and Triton kernels for GPU acceleration. Similar to Xiao et al [22], we adopt depthwise spatial convolutions and pointwise channel mixing in residual spatial-mixing blocks to aggregate spatial neighbourhood information and fuse information between channels. These spatial-mixing blocks are applied before, between, and after the two sequence-core passes. An additional residual depthwise 1D convolution supplies local, non-causal depth context. Motivated by previous embeddings of the beam energy to condition proton dose predictions [8, 9, 15], the proton model additionally includes a learned per-energy embedding which is prepended as a prefix token to the Mamba temporal sequence. For photon dose prediction, a second Mamba-3 branch scans the reversed depth sequence and is fused with the forward branch through a learnable, zero-initialised gate, enabling the model to predict dose contributions from material downstream of it, principally scatter, which a causal core cannot represent.

Decoder and CT space reconstruction. The latent features from depth-sequence modelling are reshaped into depth-indexed feature grids and decoded slice-wise in 3 transposed-convolution stages with LeakyReLU or ReLU activation, followed by a 1×1 convolution producing four packed coefficient channels using sub-pixel upsampling [18]. The head predicts four phase-packed cubic B-spline coefficient channels [20], representing a two-fold finer dose field along depth and height without explicitly materialising a high-resolution BEV volume. This is performed to reduce penumbra smearing and artifacts when transforming BEV dose to CT space. For the proton model only, the coefficients are subsequently refined by a residual convolutional head conditioned on packed remaining range, whereas the photon coefficients are passed directly to the resampler. A custom differentiable Triton kernel performed fast evaluation of the continuous spline field at patient CT voxel centres using the inverse BEV-to-CT transformation.

Synthetic CT Generation For magnetic resonance imaging (MRI) tasks, MRI was converted to sCT using a cGAN [3]. The generator was a SwinUNETR with feature size 48 [2], and the discriminator was a 3D PatchGAN. The generator operated on $64 \times 160 \times 160$ patches. MRI volumes were standardised per subject, CT volumes using training-population statistics, and a body mask restricted optimisation to relevant anatomy. The resulting sCT was passed to the corresponding CT-based dose model.

Parameters and initialisation. The photon and proton dose models contain 1.77 M and 0.73 M trainable parameters, respectively; the sCT cGAN contains 75.41 M parameters (72.76 M generator; 2.64 M discriminator). Convolutional and linear layers used PyTorch’s default Kaiming-uniform initialisation. The photon model was warm-started from a unidirectional checkpoint, with its backward-fusion gate zero-initialised, while the range-conditioned proton model was warm-started from a model trained without the Bragg residual head. The initial dose checkpoints and both sCT models were trained from random initialisation. No external pretrained weights were used.

Models were implemented in PyTorch using mamba-ssm 2.3.2 for the Mamba-3 cores and Triton 3.5.0 for custom packed-spline evaluation and BEV-to-CT resampling kernels. SimpleITK handled medical-image I/O and spatial transformations, while MONAI 1.6.0 provided data-loading components and the SwinUNETR sCT generator. Source code, complete package versions, and model configurations will be made available at <https://github.com/doserad-IX/ImageX-DoseRAD2026>.

2.3 Training

Dose-model training. CT values were clipped to $[-1024, 3071]$ and rescaled to $[0, 1]$. Reference doses were normalised using a cohort-specific scalar estimated from the training patients and restored to physical units during inference. No explicit data augmentation was applied to the dose models. Beam-specific BEV inputs were generated on-the-fly from the treatment geometry.

We used a staged learning approach for our dose prediction models to accelerate model convergence, with the added benefit of being able to learn primary transport physics before learning secondary residual corrections. Photon training comprised of three stages. First, a unidirectional model was trained from random initialisation on the thorax cohort for 200 epochs using an initial learning rate of 2.5×10^{-4} . This checkpoint was then fine-tuned on the combined thorax and abdomen training cohorts for 60 epochs using an initial learning rate of 1×10^{-4} . Finally, the reverse-depth Mamba branch was added and the resulting bidirectional model was fine-tuned for a further 100 epochs, also with an initial learning rate of 1×10^{-4} . The proton model was trained only on the thorax training cohort from random initialisation on the thorax cohort for 200 epochs without the Bragg residual head. The residual head was then added and the complete model fine-tuned for 150 epochs. Both proton stages used an initial learning rate of 1×10^{-4} .

Both dose models minimised CT-space mean squared error (MSE) between the predicted and reference dose distributions and included a masked MAE term over voxels receiving at least 10% of the reference maximum. The photon objective used a masked-MAE weight of 2×10^{-4} :

$$\mathcal{L}_{\text{photon}} = \text{MSE}(\hat{y}, y) + 2 \times 10^{-4} \text{mean}_{y \geq 0.1 \max(y)} |\hat{y} - y|. \quad (4)$$

The proton objective used a masked-MAE weight of 1×10^{-4} and additionally penalised negative predictions within the valid sampling region:

$$\begin{aligned} \mathcal{L}_{\text{proton}} = & \text{MSE}(\hat{y}, y) + 1 \times 10^{-4} \text{mean}_{y \geq 0.1 \max(y)} |\hat{y} - y| \\ & + 0.01 \text{mean}_{\text{valid}} [\max(0, -\hat{y})^2]. \end{aligned} \quad (5)$$

The photon models were optimised using:

$$\mathcal{L}_{\text{photon}} = \text{MSE}(\hat{y}, y) + \lambda_{\text{MAE}} \cdot \text{MAE}(\hat{y} \odot M, y \odot M), \quad (6)$$

where $\text{MSE}(\hat{y}, y)$ is the mean-squared-error between predicted ($\hat{y}$) and reference (y) dose distributions, λ_{MAE} is the mean absolute error (MAE) weighting factor (set to 2×10^{-4}), $\text{MAE}(\hat{y}, y)$ is the mean absolute error, and M is a threshold mask generated by $y \geq 0.1 \cdot \max(y)$.

The proton models were optimised using:

$$\mathcal{L}_{\text{proton}} = \text{MSE}(\hat{y}, y) + \lambda_{\text{MAE}} \cdot \text{MAE}(\hat{y} \odot M, y \odot M) + \lambda_{\text{valid}} [\max(0, -\hat{y}^2)], \quad (7)$$

where λ_{MAE} was set to 1×10^{-4} , λ_{valid} is the valid loss weighting factor (set to 1×10^{-2}), and $[\max(0, -\hat{y}^2)]$ penalised negative predictions within the valid sampling region.

All dose models were trained using Adam with batch size 4, no weight decay, and global gradient-norm clipping at 1.0, with four data-loader workers for the photon runs and eight for the proton runs. The learning rate was halved when validation loss did not improve for 10 photon or 15 proton epochs. Training used bfloat16 mixed precision.

Synthetic CT training. Following the SynthRAD2025 KoalAI (winning MRI to CT model submission) preprocessing approach [17, 19], where the MRIs were standardised using a subject z-score while the CTs were standardised using a population mean and standard deviation, enabling the recovery of HU by inverse normalisation based on these population-level statistics. A body mask restricted optimisation to anatomically relevant voxels. The sCT cGAN was trained on $64 \times 160 \times 160$ patches for up to 1000 epochs. Data augmentation comprised of random cropping and flipping, intensity and contrast perturbations, and additive Gaussian noise. The sCT model was optimised using:

$$\mathcal{L}_{\text{G,total}} = \lambda_{\text{G,image}} \cdot \mathcal{L}_{\text{G,image}} + \mathcal{L}_{\text{G,adv}}, \quad (8)$$

where $\lambda_{\text{G,image}}$ is the generator image content weighting term (set to 100), $\mathcal{L}_{\text{G,image}}$ is the generator image content loss term (MSE), and $\mathcal{L}_{\text{G,adv}}$ is the adversarial loss term (MSE, based on least-squares GAN [12]). The generator

and discriminator were optimised using AdamW [11] and Adam [5], respectively. The checkpoint with the lowest validation HU MAE was retained.

Hyperparameter and model selection. Hyperparameters were selected through targeted experiments on the held-out patients described in Section 2.1. Within each training stage, the checkpoint with the lowest validation loss was retained. Candidate model configurations were compared using the official beam-level metrics on the held-out validation set. No model ensembling was used, which makes for efficient runtime.

2.4 Evaluation

During training, MSE and masked MAE were monitored on the held-out validation patients. Beam-level performance was evaluated using the challenge-provided code for: (i) masked MAE, calculated over voxels receiving at least 10% of the reference beam maximum and normalised by that maximum; and (ii) integrated depth-dose (IDD) distance, defined as the normalised root-mean-square difference between predicted and reference IDD curves along the beam direction.

Beam-level results were averaged within each patient and summarised across patients using the mean and standard deviation. Individual patient values are also reported. Given the small validation cohorts, no confidence intervals or formal significance tests were calculated.

Packaged models were additionally evaluated on the hidden test set through the DoseRAD2026 Grand Challenge platform. Official plan-level metrics were: (i) stratified MAE, averaged across low-dose (10–30%), intermediate-dose (30–80%), and high-dose ($\geq 80\%$) regions; (ii) the 3D local gamma pass rate at 1%/1 mm; and (iii) a DVH-based clinical score combining relative errors in PTV $D_{98\%}$ and $V_{95\%}$ and in $D_{2\%}$ and mean dose for the three organs-at-risk closest to the target.

Runtime was estimated by the organisers using a non-negative linear model of fixed, per-image, and per-dose-map costs. The fitted model was evaluated for representative cases comprising one image and either 181 photon control points or 500 proton beamlets.

No formal model-explainability method was applied. Depth-resolved IDD curves and spatial dose-error maps were instead inspected as qualitative diagnostics of systematic dose errors, including dose mismatch and penumbra misalignment.

3 Results

Challenge performance Table 2 summarises performance on the DoseRAD2026 preliminary test set. The CT-based photon and proton models achieved similar beam-level accuracy, with masked MAE of 0.0084 for both and local gamma pass rates of 95.6% and 95.4%, respectively. The corresponding MRI-based tasks produced higher errors, particularly for proton dose prediction: relative to CT-to-proton prediction, MRI-to-proton masked MAE increased from 0.0087 to 0.0326,

while the gamma pass rate decreased from 95.4% to 75.4%. The substantially larger degradation for MRI-to-proton prediction highlights sCT accuracy as a major limitation for MRI-based proton dose calculation.

Table 2. Evaluation metrics for CT/MRI to photon/proton dose calculated on the preliminary test set for the DoseRAD2026 challenge. MAE: beam-level mean absolute dose error; IDD Dist.: normalised integrated depth-dose curve distance; Plan MAE: stratified plan-level MAE; γ Index: 3D local gamma pass rate (%) at 1%/1 mm; DVH Score: DVH-based clinical score.

Task	MAE	IDD Dist.	Plan MAE	γ Index	DVH Score	Runtime
CT to photon	0.0084	0.0069	0.0041	96.7569	0.2173	23.4493
	± 0.0022	± 0.0105	± 0.0019	± 3.0172	± 0.0663	
MRI to photon	0.0139	0.0097	0.0093	89.7046	0.9051	39.6613
	± 0.0064	± 0.0093	± 0.0056	± 7.2553	± 0.1782	
CT to proton	0.0084	0.0058	0.0079	95.9530	0.5010	22.5295
	± 0.0039	± 0.0081	± 0.0039	± 3.6107	± 0.2955	
MRI to proton	0.0326	0.0287	0.0336	75.4368	9.5124	42.8198
	± 0.0216	± 0.0208	± 0.0226	± 18.9888	± 9.5922	

Ablation experiments Table 3 summarises pairwise development experiments assessing BEV extent, model capacity, sequence core, depth directionality, CT-space training, grid snapping, and phase packing. For each comparison, the lowest-validation-loss checkpoint within the epoch range shared by both runs was used. Because experiments were conducted at different development stages, the cohort and secondary metric vary by comparison; results should therefore be interpreted within, rather than across, pairs.

All pairwise comparisons favoured the adopted configuration. On the thorax cohort, increasing BEV depth from 256 to 384 planes reduced masked MAE by 8.5% and IDD distance by 30.0%, while doubling model capacity reduced masked MAE by 15.5%. Adding the reverse-depth Mamba branch reduced masked MAE by 5.3% and the IDD distance by 20.9%. Pilot experiments also favoured CT-space training with axial grid snapping, implicit super-resolution inputs and outputs to reduce interpolation error between BEV and CT-space. Several pairs differed in additional settings, so these results indicate associations rather than isolated component effects.

Qualitative error analysis Figure 2 compares predicted and reference dose for photon beams and proton beamlets with high IDD error and were selected to illustrate failure modes. The prediction reproduced the overall beam trajectory while the largest discrepancies occurred at peak dose deposition. The corresponding IDD curves showed the mismatch between ground truth and predicted doses and a range shift for the proton case which is consistent with the spatial difference map. For the MRI-based tasks, (b) shows an visible striation artifacts in the sCT. In both (b) and (d) inaccurate soft tissue and bone HU coincided with local dose errors.

Table 3. Experiments motivating design decisions final model. Each pair varies the stated factor; the epoch is the lowest-validation-loss epoch within the range common to both runs. Metric sets differ by experiment generation: ^a BEV-space masked MAE and BEV IDD distance; ^b CT-space masked MAE and round-trip BEV IDD distance; ^c CT-space masked MAE and true-gantry beam IDD distance; ^d CT-space masked MAE and CT-space MSE ($\times 10^{-6}$), the two runs sharing no IDD variant for the reason given in the text. Rows marked † vary one further factor and ‡ several (see text). Lower is better throughout. SR: super-resolution.

Design decision	Variant	Epoch	MAE	IDD
<i>Thorax cohort, 36 training patients</i>				
BEV depth extent ^b	256 planes	122	0.0117	0.0120
	384 planes	124	0.0107	0.0084
Model capacity ^b	1.0×	94	0.0116	0.0076
	2.0×	96	0.0098	0.0070
Sequence core ^{b,c} ‡	mLSTM	115	0.0104	–
	Mamba-3	138	0.0090	–
<i>Merged cohort, warm-started from the unidirectional run</i>				
Depth directionality ^c	unidirectional	72	0.0075	0.0043
	bidirectional	75	0.0071	0.0034
<i>Pilot cohort, 10 training patients, thorax and abdomen</i>				
CT-space training	BEV-space training	100	0.0195	–
	CT-space + SR	100	0.0132	–
Axial grid snapping ^a †	off	60	0.0150	0.0135
	on	65	0.0140	0.0125
Super-sampled inputs ^d	off	68	0.0141	–
	on	66	0.0138	–

4 Discussion

Performance across input modalities. The CT-based photon and proton models achieved similar performance, with local gamma pass rates of approximately 96% at 1%/1 mm and normalised masked MAE below 0.9% of the reference beam maximum. Performance decreased when planning CT was replaced by sCT, with a substantially larger reduction for proton than photon prediction. Relative to the corresponding CT task, the MRI-to-photon gamma pass rate decreased by 7.3%, whereas the MRI-to-proton rate decreased by 21.4%.

The greater proton sensitivity is consistent with the dependence of proton range on integrated stopping power. HU errors along a beam path can shift the predicted Bragg peak, whereas comparable errors generally perturb photon attenuation without producing the same depth displacement. In addition, sCT error affects the proton model both through the image and through the WET and remaining-range volumes derived from it. The qualitative examples show spatial coincidence between sCT inaccuracies and dose discrepancies, but do not establish causality.

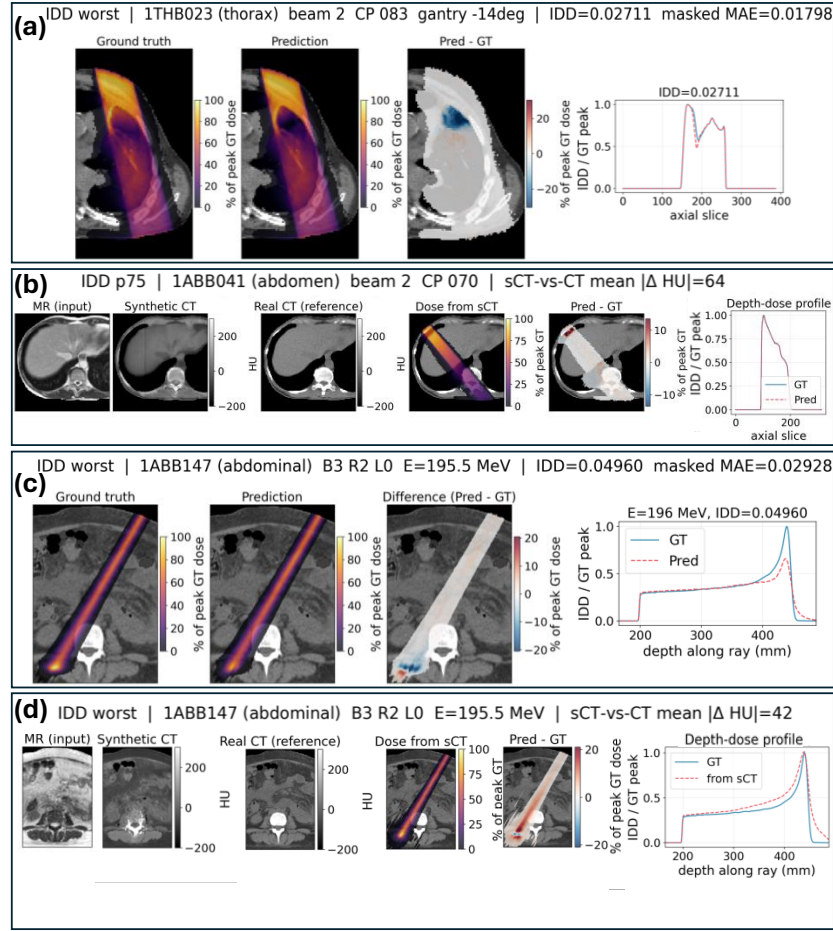


Fig. 2. Failure modes in photon and proton dose estimates taken from the held out validation set. (a) and (c) show a comparison of ground truth, prediction, difference dose and IDD curve prediction for a CT input to photon beam prediction and a proton beamlet prediction respectively. (b) and (d) show a comparison of the MR input, sCT, real CT, prediction and dose difference for photon beam and a and proton beamlet respectively. (d) also includes IDD curves to visualise the range shift and dose dropoff mismatch.

Architectural design choices. The development experiments favoured increased greater model representational capacity: BEV extent, increased model width, and bidirectional depth modelling. Grid snapping along the S-I direction and CT-space training utilising super-sampling over the remaining spatial dimensions (both CT inputs and spline coefficient outputs) enabled significant accuracy improvement via reduced interpolation error. Negligible impact on inference time was achieved through phase-packing the super-resolved channels alongside dedi-

cated Triton kernels for the resampling. The largest observed improvements were associated with this supersampling, BEV extent, model width, and bidirectional depth processing. In particular, the reverse-depth Mamba branch improved IDD distance more than masked MAE, consistent with its intended role in modelling depth-dependent effects that are unavailable to a strictly forward scan. However, the experiments used different cohorts and metric implementations, and several comparisons varied additional settings; their effect sizes are therefore not directly comparable or attributable to single components.

Limitations. The validation cohorts contained only two or three patients per task, preventing reliable confidence intervals or formal patient-level significance testing. While the development experiments provided directional insights motivating our architectural choices, they were conducted iteratively under strict challenge deadlines and computation constraints and the design of the ablation study could be improved to strictly isolate a single variable and use consistent evaluation metric definitions. The temporal convolution, energy prefix, slice scaler, and Bragg residual head were not independently ablated. Due to compute and time constraints, proton fine-tuning was not completed on the full training cohort, potentially limiting anatomical generalisability. The sCT model performance may also depend on scanner and acquisition characteristics, and the simplified RSP and range approximations may not generalise beyond the energies and anatomy represented in the challenge dataset.

5 Author Contributions

D.E.J.W. — Writing – review & editing, Supervision & Resources

E.H. — Writing – review & editing, Supervision

C.C. – Conceptualisation, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualisation

M.F. – Conceptualisation, Methodology, Software, Validation, Formal Analysis, Investigation, Writing – Original Draft, Visualisation

J.G. – Conceptualisation, Methodology, Software, Validation, Investigation, Writing – Original Draft

6 Other Information

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Disclosure of Interests. None to declare.

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